

2025

COMPUTER SCIENCE — HONOURS — PRACTICAL

Paper : DSCC-8P

(Object Oriented Programming using Java)

Full Marks : 25

Set - 5

Marks Distribution

Source Code :	10
Output :	05
Laboratory Notebook :	05
Viva voce :	05
	25

Answer *any one* question.

1. Create a Java class named MyPoint to represent a point in 2D space with x and y coordinates. The class should include the following :

Two instance variables : x and y, both of type int.

- A **default constructor** that initializes the point to the origin (0, 0).
- A method setXY(int x, int y) to update both coordinates.
- A method getXY() that returns the x and y values as a two-element integer array.
- A toString() method that returns the point in the format "(x, y)".
- A method distance(int x, int y) that calculates and returns the distance from the current point to another point with the given coordinates.

Also, write a **test class** named TestMyPoint to create instances of MyPoint and test all its public methods.

(2)

D(4th Sm.)-Computer Science-HIPr.IDSCC-8P/CCF(Set-5)

2. Write a program in Java that accepts a square matrix and check whether the matrix is symmetric matrix or not.

A square symmetric matrix is a square matrix that is equal to its transpose, i.e. $A=A^T$

Input : 1 2 3

2 4 5

3 5 6

Output : Symmetric Matrix.

3. Write a Java program that implements the binary search algorithm to find a target element in a given array of integers.
4. Write a Java program to sort a list of numbers using Selection sort and display its result. User must send the input as command line argument.
5. Write a Java program that calculates the degree of a specific node in an undirected graph. The degree of a node is the number of edges connected to it.
6. Write a Java program to classify a given array of integers if it contains duplicate element. Store the classification result in a variable called result. If the result is False, then calculate the average value of the array. Else, remove the duplicate elements, print the unique elements in the array and then calculate the average value of the array.
7. Write a Java program to create your own exception class called InvalidAgeException that checks if you are eligible to caste vote. This exception will be thrown when entered age is below 18.